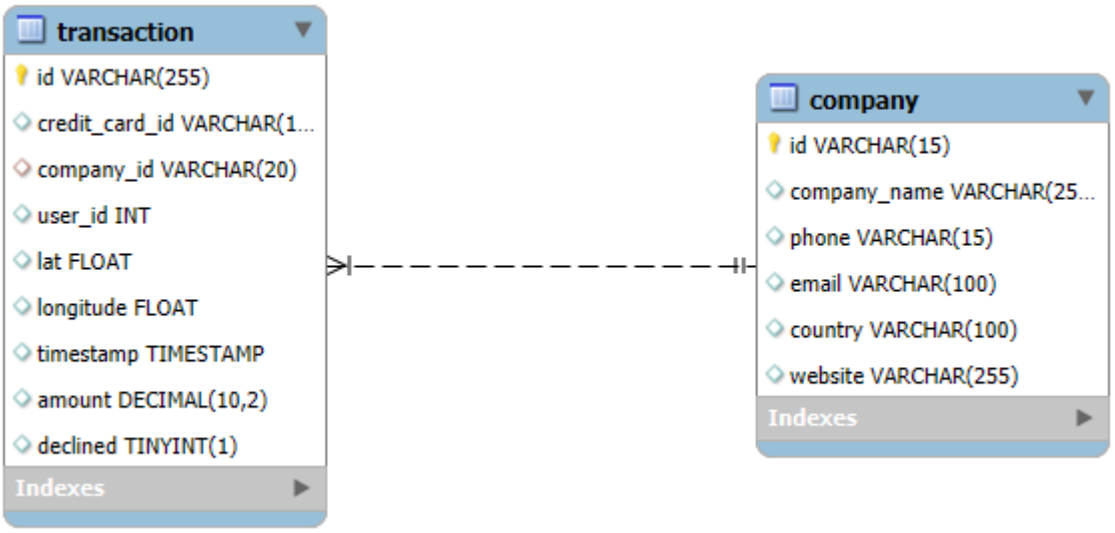
NIVEL 1 - EXERCICI 1

Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.



CARDINALIDAD

La tabla transaction está vinculada con la tabla company en **una relación de muchos a uno** (M:1). Diferentes filas ubicadas en la tabla transaction pueden estar vinculadas a una sola fila de la tabla company.

CLAVES

Las tablas transaction y company contienen sus **PRIMARY KEYS** respectivas que **identifican de forma única** cada una de sus transacciones y compañías. La tabla transaction contiene la **FOREIGN KEY** company_id mediante la cuál identifica de forma única la **PRIMARY KEY** id de company permitiendo la unión de ambas tablas dentro del esquema.

NIVEL 1 - EXERCICI 2

```sql
DESCRIBE transaction;
DESCRIBE company;
SELECT * FROM transaction ORDER BY RAND() LIMIT 100;
SELECT * FROM company ORDER BY RAND() LIMIT 100;
-- ####################
-- NIVEL 1 ############
-- ####################
-- Exercici 2 #########
-- ####################

-- Llistat dels països que estan generant vendes.

SELECT DISTINCT c.country
FROM transaction t
JOIN company c ON c.id = t.company_id
WHERE t.declined = 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country
Netherlands
Sweden
Ireland
United States
Belgium
Canada
Germany
Norway

Result 21

Output

Action Output

#	Time	Action	Message
1	10:40:13	SELECT DISTINCT c.country FROM transaction t JOIN company c ON c.id = t.company_id WHERE t.declined = 0	15 row(s) returned

```sql
-- Des de quants països es generen les vendes.

SELECT COUNT(DISTINCT c.country)
FROM transaction t
join company c on c.id = t.company_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(DISTINCT c.country)
15

Result 22

Output

Action Output

#	Time	Action	Message
1	10:42:55	SELECT COUNT(DISTINCT c.country) FROM transaction t join company c on c.id = t.company_id	1 row(s) returned

```sql
-- Identifica la companyia amb la mitjana més gran de vendes.

SELECT c.company_name,
        ROUND(AVG(t.amount),2) AS ventas
FROM transaction t
JOIN company c ON c.id = t.company_id
GROUP BY c.company_name
ORDER BY ROUND(AVG(t.amount),2) DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	ventas
Ac Fermentum Incorporated	284.87

Result 24

Output

Action Output

#	Time	Action	Message
1	10:44:27	SELECT c.company_name, ROUND(AVG(t.amount),2) AS ventas FROM transaction t JOIN company c ON c.id = t.company_id GROUP BY c.com...	1 row(s) returned

NIVEL 1 - EXERCICI 3

```
-- ####################
-- NIVEL 1 ############
-- ####################
-- Exercici 3 #########
-- ####################

-- Utilitzant només subconsultes (sense utilitzar JOIN):

-- Mostra totes les transaccions realitzades per empreses
-- d'Alemanya.

SELECT t.id
FROM transaction t
WHERE EXISTS (
                    SELECT c.id
                    FROM company c
                    WHERE t.company_id = c.id
                    AND c.country = "Germany"
                );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id
00138D3B-206D-4C03-94B7-63A2676EB9B4
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9
00201A11-2E62-44C4-941D-198FC8DB77F0
00235618-0A5C-4D49-9DCB-B3A9405D8923
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38
00687139-48B2-4FFA-8E73-B20376F04AB4
0074F4DD-32F1-4827-8758-55896314623A
00AAB9CD-39D6-4DCB-8A1D-13BE73DC90A9

transaction 26

Output

Action Output

#	Time	Action	Message
1	10:48:52	SELECT t.id FROM transaction t WHERE EXISTS (SELECT c.id FROM company c WHERE t.company_id = c.id AND c....	13291 row(s) returned

```sql
-- Llista les empreses que han realitzat transaccions
-- per un amount superior a la mitjana de totes les transaccions.

SELECT DISTINCT c.company_name
FROM company c
WHERE EXISTS (
            SELECT t.company_id
            FROM transaction t
            WHERE t.company_id = c.id
            AND t.amount > (
                            SELECT AVG(t.amount)
                            FROM transaction t
                            WHERE t.declined = 0
                            )
            AND t.declined = 0
            );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name
Ac Fermentum Incorporated
Magna A Neque Industries
Fusce Corp.
Convallis In Incorporated
Ante Iaculis Nec Foundation
Donec Ltd
Sed Nunc Ltd
Amet Nulla Donec Corporation

company 28

Output

Action Output

#	Time	Action	Message
1	10:49:42	SELECT DISTINCT c.company_name FROM company c WHERE EXISTS (SELECT t.company_id FROM transaction t WHERE t.compan...	100 row(s) returned

```sql
-- Eliminaran del sistema les empreses que no tenen transaccions registrades,
-- entrega el llistat d'aquestes empreses.

SELECT DISTINCT c.company_name
FROM company c
WHERE NOT EXISTS (
                SELECT t.company_id
                FROM transaction t
                WHERE t.company_id = c.id
                );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name

company 30

Output

Action Output

#	Time	Action	Message
1	10:50:08	SELECT DISTINCT c.company_name FROM company c WHERE NOT EXISTS (SELECT t.company_id FROM transaction t WHERE t....	0 row(s) returned

NIVEL 2 - EXERCICI 1

```sql
-- ####################
-- NIVEL 2 ############
-- ####################
-- Exercici 1 #########
-- ####################

-- Identifica els cinc dies que es va generar la quantitat més gran
-- d'ingressos a l'empresa per vendes. Mostra la data de cada transacció
-- juntament amb el total de les vendes.


SELECT CAST(t.timestamp AS DATE) AS dia,
        c.company_name,
        ROUND(SUM(t.amount),2) AS ventas
FROM transaction t
JOIN company c on c.id = t.company_id
WHERE t.declined = 0
GROUP BY dia, c.company_name
ORDER BY ventas DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dia	company_name	ventas
2024-12-02	Eget Ipsum Ltd	3398.40
2019-03-19	Ac Fermentum Incorporated	3149.04
2017-12-20	Nulla Integer Vulputate Corp.	2774.04
2017-12-20	Viverra Donec Foundation	2773.35
2020-12-11	Ac Fermentum Incorporated	2717.76

Result 32

Output

Action Output

#	Time	Action	Message
1	10:50:41	SELECT CAST(t.timestamp AS DATE) AS dia, c.company_name, ROUND(SUM(t.amount),2) AS ventas FROM transaction t JOIN company c ...	5 row(s) returned

```sql
-- Quina és la mitjana de vendes per país?
-- Presenta els resultats ordenats de major a menor mitjà.

SELECT c.country, ROUND(AVG(t.amount),2) AS media_de_ventas
FROM company c
JOIN transaction t ON t.company_id = c.id
GROUP BY country
ORDER BY media_de_ventas DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	media_de_ventas
Australia	265.19
United States	264.98
Belgium	261.15
Germany	260.84
Ireland	260.64
Spain	260.47
France	259.98
New Zealand	259.59

Result 34

Output

Action Output

#	Time	Action	Message
1	10:51:27	SELECT c.country, ROUND(AVG(t.amount),2) AS media_de_ventas FROM company c JOIN transaction t ON t.company_id = c.id GROUP BY cou...	15 row(s) returned

```
-- ####################
-- NIVEL 2 ############
-- ####################
-- Exercici 2 #########
-- ####################

-- En la teva empresa, es planteja un nou projecte per a llançar algunes
-- campanyes publicitàries per a fer competència a la companyia "Non Institute".
-- Per a això, et demanen la llista de totes les transaccions realitzades per
-- empreses que estan situades en el mateix país que aquesta companyia.

-- Mostra el llistat aplicant JOIN i subconsultes.

SELECT t.id
FROM transaction t
JOIN company c ON c.id = t.company_id
WHERE t.declined = 0
AND c.country = (
                    SELECT DISTINCT c.country
                    FROM company c
                    JOIN transaction t ON t.company_id = c.id
                    WHERE c.company_name = "Non Institute"
                    )
AND c.company_name <> "Non Institute";
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id
008629B4-C9A9-406C-A3D2-71FDA47BC546
00B72BA4-54A3-4B8E-B13F-2D57535AA17A
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C
023FFCE8-E618-4938-BF56-C8DF80540ADD
02683BEB-EF91-4564-957B-D6F1662AB7C5
02C2F29E-CEF2-4C1E-A594-F476E8F279C0
02F468DC-426C-47C2-8B0A-D8B25B7A81AF
03068F3B-817B-4A49-934F-0F439291A104

Result 35

Output

Action Output

#	Time	Action	Message
1	10:52:04	SELECT t.id FROM transaction t JOIN company c ON c.id = t.company_id WHERE t.declined = 0 AND c.country = (SELECT DISTINCT c.country ...	12213 row(s) returned

```
142 •  SELECT t.id
143    FROM transaction t
144    WHERE t.declined = 0
145    AND EXISTS (
146                        SELECT 1
147                        FROM company c
148                        WHERE c.id = t.company_id
149                        AND c.country = (
150                                    SELECT DISTINCT c.country
151                                    FROM company c
152                                    WHERE c.company_name = "Non Institute")
153                                            )
154    AND NOT EXISTS (
155                            SELECT 1
156                            FROM company c
157                            WHERE c.id = t.company_id
158                            AND c.company_name = "Non Institute"
159                            );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id
008629B4-C9A9-406C-A3D2-71FDA47BC546
00B72BA4-54A3-4B8E-B13F-2D57535AA17A
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C
023FFCE8-E618-4938-BF56-C8DF80540ADD
02683BEB-EF91-4564-957B-D6F1662AB7C5
02C2F29E-CEF2-4C1E-A594-F476E8F279C0
02F468DC-426C-47C2-8B0A-D8B25B7A81AF
0306BF3B-817B-4A49-934F-0F439291A104

transaction 38

Output

Action Output

#	Time	Action	Message
1	10:54:22	SELECT t.id FROM transaction t WHERE t.declined = 0 AND EXISTS (SELECT 1 FROM company c WHERE c.id = t.company_id A...	12213 row(s) returned

NIVEL 3 - EXERCICI 1

```sql
-- ####################
-- NIVEL 3 ############
-- ####################
-- Exercici 1 #########
-- ####################

-- Presenta el nom, telèfon, país, data i amount,
-- d'aquelles empreses que van realitzar transaccions amb
-- un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates:
-- 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024.
-- Ordena els resultats de major a menor quantitat.

SELECT c.company_name,
        c.phone,
        c.country,
        CAST(t.timestamp AS DATE) AS fecha,
        ROUND(t.amount,2) AS cantidad
FROM transaction t
JOIN company c ON c.id = t.company_id
WHERE (t.amount BETWEEN 350 AND 400)
AND CAST(t.timestamp AS DATE)= "2015-04-29"
    OR CAST(t.timestamp AS DATE) = "2018-07-13"
    OR CAST(t.timestamp AS DATE) = "2024-03-13"
ORDER BY cantidad DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	country	fecha	cantidad
Justo Eu Arcu Ltd	08 42 56 71 52	Italy	2018-07-13	517.65
Maecenas Malesuada Fringilla ...	09 38 53 76 61	Netherlands	2018-07-13	496.12
Tristique Neque Venenatis Ins...	04 34 85 12 85	Sweden	2018-07-13	485.15
Integer Mollis Corp.	03 12 20 45 24	Italy	2024-03-13	473.83
Non Ante LLP	08 89 47 65 08	Sweden	2018-07-13	449.52
Non Ante LLP	08 89 47 65 08	Sweden	2018-07-13	449.52
Nunc Ac PC	03 53 62 76 95	Sweden	2018-07-13	445.95
Mauris Institute	05 29 60 36 87	Sweden	2018-07-13	445.43

Result 39

Output

Action Output

#	Time	Action	Message
1	10:54:43	SELECT c.company_name, c.phone, c.country, CAST(t.timestamp AS DATE) AS fecha, ROUND(t.amount,2) AS cantidad FROM t...	38 row(s) returned

NIVEL 3 - EXERCICI 2

```sql
-- ####################
-- NIVEL 3 ############
-- ####################
-- Exercici 2 #########
-- ####################

-- Necessitem optimitzar l'assignació dels recursos i dependrà
-- de la capacitat operativa que es requereixi, per la qual cosa
-- et demanen la informació sobre la quantitat de transaccions que
-- realitzen les empreses, però el departament de recursos humans
-- és exigent i vol un llistat de les empreses on especifiquis si
-- tenen més de 400 transaccions o menys.

SELECT c.company_name,
       COUNT(t.id) AS transacciones,
    CASE
        WHEN COUNT(t.id) > 400 THEN "Si"
        ELSE "No"
    END AS mas_de_400_transacciones
FROM transaction t
JOIN company c ON c.id = t.company_id
GROUP BY c.company_name
ORDER BY transacciones DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	transacciones	mas_de_400_transacciones
Ac Fermentum Incorporated	2401	Si
Nunc Interdum Incorporated	1599	Si
Donec Fringilla PC	1593	Si
Mauris Institute	1586	Si
Rutrum Non Inc.	1585	Si
Aliquet Vel Vulputate Incorpor...	1583	Si
Aliquam PC	1576	Si
Orci Adipiscing Limited	1567	Si

Result 41

Output

Action Output

#	Time	Action	Message
1	10:55:37	SELECT c.company_name, COUNT(t.id) AS transacciones, CASE WHEN COUNT(t.id) > 400 THEN "Si" ELSE "No" END AS mas_de_400_tr...	100 row(s) returned